

A142 – Gravitational Dynamics: Time-Field Lagrange and Matter Coupling

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.1 Purpose

[STIPULATION] This document develops the complete Lagrangian formulation of gravitational dynamics in FFGFT: the intrinsic time field $\mathcal{T} = 1/\max(m, \omega)$ as the fundamental object, the field equation for spherical and extended systems, the coupling of all matter fields to the time field through conformal transformation, the modified Dirac equation, and the complete total Lagrangian density. The document thus provides the framework designated as "gravitational dynamics" in A140 and recorded there only as a bridge result. What remains open after this document is explicitly named.

.2 The Intrinsic Time Field

[B] The fundamental object of gravitational dynamics in FFGFT is not the metric but the intrinsic time field:

$$\mathcal{T}(x, t) = \frac{1}{\max(m, \omega)}$$

in natural units ($\hbar = c = 1$). $[\mathcal{T}] = [E^{-1}]$. The field is reciprocal to the local energy – heavy objects have small $\mathcal{T}$ (rapidly oscillating), light ones have large $\mathcal{T}$. The time-mass duality $\tilde{T} \cdot m = 1$ (A020/A060) is the zero-field equation of this structure.

[B] Three fundamental field geometries:

Geometry	Parameters
Localised spherical	$\beta = 2Gm/r$ (dimensionless), $\xi = 2\sqrt{G} \cdot m$ (dimensionless), $\mathcal{T}(r) = (1/m_0)(1 - \beta)$
Localised non-spherical	$\beta_{ij} = r_{0ij}/r$ (tensor), $\xi_{ij} = 2\sqrt{G} \cdot I_{ij}$
Infinite homogeneous	$\nabla^2 m = 4\pi G \rho_0 m + \Lambda_T m$, $\xi_{\text{eff}} = \xi/2$

[K] For quantum computing applications and atomic scales ($R/r \gg 10^{12}$) only the localised spherical geometry is relevant; the others are for theoretical completeness.

.3 The Field Equation

[B] For a point-mass source $\rho = m\delta^3(\mathbf{r})$ the complete geometric solution is:

$$m(r) = m_0 \left(1 + \frac{2Gm}{r}\right) = m_0(1 + \beta), \quad \mathcal{T}(r) = \frac{1}{m_0}(1 + \beta)^{-1} \approx \frac{1}{m_0}(1 - \beta).$$

The factor 2 in $r_0 = 2Gm$ follows from the relativistic field structure and coincides exactly with the Schwarzschild radius. The field equation:

$$\nabla^2 m = 4\pi G \rho(x, t) \cdot m. \quad [E^3 = E^{-2} \cdot E^4 \cdot E = E^3] \checkmark$$

For infinite systems with background consistency:

$$\nabla^2 m = 4\pi G \rho_0 m + \Lambda_T m, \quad \Lambda_T = -4\pi G \rho_0.$$

The Λ_T term generates the cosmic shielding effect $\xi_{\text{eff}} = \xi/2$.

[B] No new objects. Neither $r_0 = 2Gm$ nor $\Lambda_T = -4\pi G \rho_0$ are independent quantities beside the scale structure from ξ . r_0 is the characteristic length of *the respective mass* – the same reciprocity as L_0 , only with m instead of the Planck mass as reference. Λ_T is the same field equation, computed from the homogeneous background instead of the point source. Both are reference-point choices, not new physics – the same logic as the reference-point table in A262.

.4 The Universal Scale Parameter from Higgs Matching

[K] The fundamental scale parameter ξ is uniquely determined by the Higgs physics:

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.33 \times 10^{-4}$$

with Higgs self-coupling $\lambda_h \approx 0.13$, Higgs VEV $v \approx 246$ GeV, Higgs mass $m_h \approx 125$ GeV. Dimensional verification: $[\xi] = [\lambda_h^2 v^2]/[16\pi^3 m_h^2] = [E^2]/[E^2] = [1]$. The condition $\beta_T = 1$ in natural units ($\hbar = c = \alpha_{\text{EM}} = \beta_T = 1$) fixes ξ uniquely and eliminates all free parameters.

.5 The Lagrangian Formulation

[B] The time-field Lagrangian density:

$$\mathcal{L}_{\text{time}} = \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \mathcal{T} \partial_\nu \mathcal{T} - V(\mathcal{T}) \right].$$

Dimensional consistency: $[\mathcal{L}_{\text{time}}] = [E^{-4}][E^2] + [E^4] = [E^0]$.

[B] Matter field coupling through conformal transformation. All matter fields couple to the time field via:

$$g_{\mu\nu} \rightarrow \Omega^2(\mathcal{T}) g_{\mu\nu}, \quad \Omega(\mathcal{T}) = \frac{\mathcal{T}_0}{\mathcal{T}} \quad [\text{dimensionless}].$$

Scalar fields:

$$\mathcal{L}_\phi = \sqrt{-g} \Omega^4(\mathcal{T}) \left(\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m^2 \phi^2 \right). \quad [\mathcal{L}_\phi] = [E^{-4}][1][E^4] = [E^0] \checkmark$$

Fermion fields:

$$\mathcal{L}_\psi = \sqrt{-g} \Omega^4(\mathcal{T}) (i \bar{\psi} \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi). \quad [\mathcal{L}_\psi] = [E^{-4}][1][E^4] = [E^0] \checkmark$$

Higgs time-field coupling:

$$\mathcal{L}_{\text{Higgs-T}} = |\mathcal{D}_{\text{Higgs-T}}|^2 - V(\mathcal{T}, \Phi), \quad \mathcal{D}_{\text{Higgs-T}} = \mathcal{T}(\partial_\mu + i g A_\mu) \Phi + \Phi \partial_\mu \mathcal{T}.$$

This delivers $\mathcal{T} = 1/(y\langle\Phi\rangle)$ – the connection between time field and Higgs vacuum expectation value.

[B] Complete total Lagrangian density:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{time}} + \mathcal{L}_{\text{gauge}} + \mathcal{L}_\phi + \mathcal{L}_\psi + \mathcal{L}_{\text{Higgs-T}},$$

with gauge term $\mathcal{L}_{\text{gauge}} = \sqrt{-g} (-\frac{1}{4} F_{\mu\nu} F^{\mu\nu})$. Each summand is dimensionally $[E^0]$.

.6 The Modified Dirac Equation

[B] In the FFGFT time-field framework the Dirac equation reads:

$$[i\gamma^\mu(\partial_\mu + \Gamma_\mu^{(T)}) - m(x, t)]\psi = 0,$$

with the time-field connection:

$$\Gamma_\mu^{(T)} = \frac{1}{\mathcal{T}} \partial_\mu \mathcal{T} = -\frac{\partial_\mu m}{m^2}.$$

This equation reduces in the limit $\mathcal{T} \rightarrow \text{const}$ ($\Gamma_\mu^{(T)} \rightarrow 0$) to the standard Dirac equation. The connection to the Schrödinger equation (A075) remains via the polar decomposition of the vacuum field.

[B] Modified Schrödinger equation:

$$i\mathcal{T} \frac{\partial \Psi}{\partial t} + i\Psi \left[\frac{\partial \mathcal{T}}{\partial t} + \mathbf{v} \cdot \nabla \mathcal{T} \right] = \hat{H} \Psi.$$

In the limit $\partial_\mu \mathcal{T} \rightarrow 0$ this reduces to the standard form $i\hbar \partial_t \Psi = \hat{H} \Psi$ with $\mathcal{T} = 1/m \leftrightarrow \hbar$.

.7 Dimensional Consistency Overview

[K]

Equation	LHS	RHS	Status
Time-field definition	$[E^{-1}]$	$[1/\max(m, \omega)] = [E^{-1}]$	$\checkmark$
Field equation	$[E^3]$	$[4\pi G \rho m] = [E^3]$	$\checkmark$
β parameter	$[1]$	$[2Gm/r] = [1]$	$\checkmark$
ξ from Higgs	$[1]$	$[\lambda_h^2 v^2 / (16\pi^3 m_h^2)] = [1]$	$\checkmark$
Lagrangian density	$[E^0]$	all terms	$\checkmark$

.8 Testable Predictions

[K] The Lagrangian framework gives specific predictions:

$$a_e^{(T0)} = \frac{\alpha}{2\pi} \cdot \xi^2 \cdot I_{\text{loop}} \approx 2.34 \times 10^{-10}$$

(QED correction to the anomalous magnetic moment). Rotation curves with modified potential: $v^2(r) = GM/r + \kappa r^2$. Energy-dependent time delay: $\Delta t = (\xi/c)(1/E_1 - 1/E_2)(2Gm/r)$.

.9 Verification Script

- Dimensional consistency of all terms numerically; limiting case Dirac $\rightarrow$ standard; ξ from Higgs matching: `python/A_Serie_Skripte/a142_gravitation_lagrange.py`

.10 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$\mathcal{T}(x, t)$	Intrinsic time field (dynamic scalar field)
Ω	Conformal transformation factor
$\Gamma_\mu^{(T)} = \partial_\mu m/m^2$	Time-field connection (connection coefficient)
$\mathcal{L}_{\text{time}}$	Lagrangian density of the time field
$G_{\mu\nu}$	Einstein tensor
$T_{\mu\nu}$	Energy-momentum tensor

.11 Open Questions

A separate quantisation of gravitation is not required in FFGFT. G is not an independent field quantity but follows intrinsically from ξ : $G = \xi^2/(4m_{\text{char}})$. Gravitation is the projection of the time-field geometry, not an independent quantum field. The question “when is gravitation quantised?” does not arise in this framework – it is already answered by the basic structure. What remains: the translation of the classical field equations into explicit loop corrections for scenarios beyond the Schwarzschild limit.

The transition to measurable predictions beyond the Schwarzschild case is not complete. The QED correction (above) is computable; predictions for strong fields, gravitational waves, or the post-Newtonian sector are not systematically worked out.

The connection to the projection chain (A090) is missing. How the gravitational sector fits into the type-I/II/III structure – whether it is a type-II loss or a type-I equivalent – is not spelled out.

.12 Supersedes

This document subsumes: Dok. 067 (time-field Lagrangian density with dimensional consistency, matter-field coupling through conformal transformations, spherically symmetric

solutions, Higgs coupling, ξ from Higgs matching, complete total Lagrangian density, modified Dirac equation, experimental predictions). *Not taken over*: the section on a static universe from Dok. 067 – it does not belong in the A-series core (A010).

.13 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.